

■ A.2.4 Counting Roots of Polynomials

The packages we have discussed so far implement algorithms that basically involve only numbers or lists of numbers. You can also write *Mathematica* programs that implement symbolic and algebraic algorithms.

Here is an example of a package that implements Sturm's method for counting the real roots of a polynomial in an interval.

```
CountRoots::usage =
  "CountRoots[f,{x,a,b}] computes the number of zeroes of the real polynomial
  f on the interval a <= x <= b. The endpoints may be infinite."

OmitZeroes[m_] := Select[m, #!=0&]

CountChanges[m_] := Block[{i,len=Length[m],num=0},
  Do[ If[m[[i]] != m[[i+1]], num++, {i,len-1} ];
  num]

CountSignChanges[m_] := CountChanges[OmitZeroes[Sign[m]]]

SturmSequence[f_,g_,x_] := Block[ { seq = {f}, p=f, q=g} ,
  While[ If[q!=0,True,False,True] ,
    AppendTo[seq,q];
    {p,q} = {q,Expand[-PolynomialRemainder[p,q,x]]};
  ];
  seq
]

SturmSequence[f_,x_] := SturmSequence[f,D[f,x],x]

CountRoots[f_,{x_,a_,b_}] := Block[{seq = SturmSequence[f,x]} ,
  CountSignChanges[ seq/.x->a ]
  - CountSignChanges[ seq/.x->b ]]
```

Counting roots of polynomials.

Read in the package.

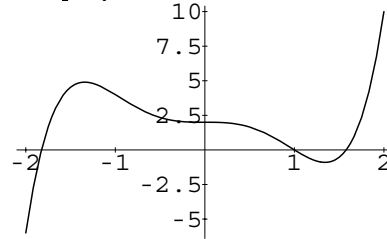
```
In[1]:= <<CountRoots.m
```

Here is a polynomial.

```
In[2]:= poly := x^5 - 3x^3 + 2
```

This shows that the polynomial has three real roots in the range -2 to 2.

```
In[3]:= Plot[poly, {x, -2, 2}]
```



You can check this result using `CountRoots`.

```
In[4]:= CountRoots[poly, {x, -2, 2}]
```

```
Out[4]= 3
```